Polynomial Arithmetic and Hensel Lifting

1. **(O2) (Hensel's Lemma)** Let $P \in \mathbb{Z}[x]$. Suppose that $p \mid P(x)$ and $p \nmid P'(x)$, where P' is the derivative of P. Prove that:
   (a) There exists $y \in \mathbb{Z}$ such that $p^2 \mid P(y)$.
   (b) For each $n \in \mathbb{N}$, there exists $y \in \mathbb{Z}$ such that $p^n \mid P(y)$.
2. **(O2)** Let $Q \in \mathbb{Z}[x]$ be a non-constant polynomial. Show that there exist arbitrarily large primes p such that $p \mid Q(n)$ for some $n \in \mathbb{Z}$.
3. **(O1)** Let $P \in \mathbb{Z}[x]$ be irreducible. Show that the quantity

   $$\gcd\big(P(n), P'(n)\big)$$

   is bounded as n ranges over $\mathbb{Z}$.
4. **(O1)** Let $P \in \mathbb{Z}[x]$.
   (a) Prove that if P has a root in $\mathbb{Z}$, then P has a root modulo p for every prime p.
   (b) Prove that if there exists a prime p such that P has no root modulo p, then P has no root in $\mathbb{Z}$.
5. **(O3)** Let $P \in \mathbb{Z}[x]$ be non-constant. Does there necessarily exist a prime p such that the sequence

   $$v_p(P(1)),\ v_p(P(2)),\ v_p(P(3)), \ldots$$

   is unbounded?